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ShareGo

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Abstract

ShareGo is a digital centralized baggage-sharing marketplace that connects travelers with unused luggage capacity to buyers demanding cheaper and efficient delivery of luggage within and across the borders. ShareGo provides an efficient, secure and cost-effective solution by resolving the problem of high courier fees, shipping costs and import taxes. It allows user to transport goods across borders by leveraging existing travel luggage space and also helps them to sell their unused luggage in thier destination area making travel easy and comfortable. Additionally, it also offers an AI chatbot to help travelers to inquire about their destination and a feature for local sales of unwanted items.

Executive Summary

In today's modern world international shipping is very difficult, particularly for small packages. It is slow and quite expensive due to the high courier fees, import taxes and customs regulations that vary from country to country. Meanwhile, millions of people travel around the world with their partially filled bags making a lot of space unused. This gap is filled by ShareGo by turning the empty luggage spaces into the practical shipping solutions. This digital platform connects the travelers with extra luggage space to people who are in search of cheaper, affordable and fast delivery options without any headache. ShareGo is a digital platform that enables travelers to make extra money by carrying goods for others, and buyers get a better deal on shipping without relying on traditional couriers and customs regulations. Buyers can request specific goods along a traveler's route, cutting out shipping costs and unwanted customs. For example, A traveler flying from China to the United States could bring an electric fan for a buyer allowing him to generate extra income by turning its unused luggage capacity into the shipping solution. Furthermore, travelers often carry items they don't really need, or they may want to free up space for something more valuable. So instead of throwing them travelers can get rid of them by simply listing the items they don't need, like the extra pair of shoes taking up space, it will allow them to earn some money from those unwanted things and will help buyer get their wanted things at low rates and better conditions. To make the experience more joyful and smoother, ShareGo integrates an AI powered chatbot to provide real-time travel assistance, such as destination insights, shopping recommendations, and local advice.

Our goal with ShareGo is to create a platform that feels secure, easy to use, cost-effective and centralized platform for everyone. Buyers can find products or request specific items from travelers headed their way, cutting down delivery time and cost in an effective manner. Travelers benefit by turning their unused luggage space into extra income while buyers benefited from cheaper and hassle-free shipping solutions.

The success of other platforms, such as Fiver and Uber, has demonstrated how widespread adoption, trust, and teamwork can transform underutilized resources into thriving economies. The marketplace, baggage sharing, and AI support all in one location is what makes ShareGo unique. Instead of just adding another layer, it is made to address the actual difficulties associated with international shipping and travel.

In conclusion, ShareGo is not about only shipping parcels or turning unused luggage space into shipping substitute; it is about making the travel experience more joyful, smoother and financially rewarding.

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Chapter 1 Introduction

Although global tourism and connectivity have increased with the advancement in technology but the movement of goods between borders still remains complex and expensive, as the customs law and duty structures varies from one country to another. At the same time millions of people travel all over the world, and most of them have unused luggage space that remains unutilized.

ShareGo is a digital baggage-sharing marketplace that connects the traveler to the user searching for cheap and affordable package movements. Also, travelers often want to get rid of items which only add few kilograms to their luggage and are of no use. This problem is also resolved by allowing users to sell these unnecessary items locally, allowing them to take financial benefits from useless items. Along with secure shipping an AI chatbot will also be integrated to enhance the travel experience with location based insights and shopping recommendations.

1.1 Purpose of this Document

In this document we will present the design, development and will evaluate the ShareGo which is our Final Year Project, which builds a digital marketplace to connect travelers with spare luggage capacity to buyers finding cheaper and quick shipping solutions. Additionally, it enables travelers to turn their unwanted goods items into financial rewards. The project addresses the challenge of expensive and complex international shipping processes by offering an alternative that makes use of underutilized travel capacity. The main goal is to create a secure, user friendly digital marketplace that lowers shipping costs and allows travelers to generate extra income, while enhancing travel experience with AI-driven insights.

1.2 Intended Audience

This main audience of this report are travelers, users seeking cheaper shipping alternatives, and developers working on digital platforms who are interested in ways to reduce the cost and improve the efficiency of international shipping. In addition, as part of the FYP requirements, this report is written for the academic audience which is mainly composed of a panel of professors looking to understand and evaluate this project.

1.3 Definitions, Acronyms, and Abbreviations

Important definitions, acronyms, and abbreviations used in this document.

FYP: Final Year Project

AI: Artificial Intelligence

UI: User Interface

UX: User Experience

API: Application Programming Interface

SDG: Sustainable Development Goal

1.4 Conclusion

In conclusion, traditional shipping is a great headache. ShareGo solves this problem by linking travelers with unused luggage space to the buyers who are finding suitable substitutes, while also enabling local item sales and offering AI-driven travel insights. This chapter describes the concept of the ShareGo that why is it important and how it fills the gap in the market.

Chapter 2 Project Vision

This chapter will provide an overview of the project problem domain, the project statement and its elaboration, the goals and objectives, and the scope of the project.

2.1 Problem Domain Overview

The domain of this project lies in international shipping and travel. Traditional shipping services are often expensive, slow, and not very flexible, while travelers frequently fly with unused luggage space. This empty space can be used to ship things from one part of the world to another providing affordable delivery solutions with the help of travelers, and travelers can also sell unwanted items locally. An AI chatbot provides travel tips such as destination insights, to improve the user experience.

2.2 Problem Statement

International shipping is slow and expensive, with high duty structures and customs delays. Many travelers have unused luggage capacity and often carry unwanted items that only adds extra kilograms to their luggage. There is no such platform that utilizes the empty luggage space for affordable shipping and help them to get rid of unwanted item while providing them financial benefits.

2.3 Problem Elaboration

In the modern era, shipping still remains a complex process. The main problem is that international shipping was not designed with regular people in mind. Smaller packages frequently cost nearly as much as larger ones, deliveries take too long, and courier charge are also high. After that customs create more frustration due to their complicated paperwork, heavy taxes and unexpected delays. Due to which a lot of people either pay excessive amounts to have small packages delivered or completely avoid sending them.

On the other hand, millions of people travel with unused luggage space which is wasted. Goods could be moved effectively and affordably in that space. Also, travelers often have unwanted items that only add few kilograms to their luggage and become a source of irritation for them. However, there is no organized way for travelers to get rid of unwanted items and to connect them with the people who need quick, affordable delivery.

This disconnect leads to a number of minor issues. First, customers miss out on quicker, less expensive shipping options. Second, passengers lose out on the opportunity to profit from unused capacity. Third,

both parties lack trust in the absence of a secure platform—no one wants to give a package to a stranger if there is no accountability or transparency.

Therefore, inefficiency, resource waste, and a lack of trust are more important factors in shipping problems than time and money. By developing a dependable system that enables buyers and travelers to locate one another, reach an agreement on terms, and safely complete deliveries, ShareGo aims to address all three.

2.4 Goals and Objectives

The main goal of ShareGo is to make international shipping and travel smarter, affordable, and cheaper by utilizing unused luggage space. The concept is to convert the unused luggage capacity into something beneficial that solves real world problems and helps people in finding good substitutes for traditional shipping.

Primary objectives of our project are:

- Create a digital marketplace where travelers can list their available luggage space along with their routes and buyers can request shipments.
- Make it easy for travelers to sell unwanted items locally, freeing up baggage space.
- Use existing travel routes rather than traditional shipping lanes to reduce delivery costs and customs hassles.
- Making the secure, cheaper, fast delivery and imports possible.
- Ensure transactions are secure and transparent so that both parties can trust the process.
- Integration of an AI chatbot to help users gain insights about destinations, and travel tips.
- Keep the UI simple, interactive and intuitive so the platform feels accessible to all kinds of users, not just to the people familiar with tech

When it comes down to all shipping solutions, creating user-friendly and reliable marketplace that work for people wanting real time solutions, ShareGo is one of them.

2.5 Project Scope

The main motive behind the making of ShareGo platform is to create a secure centralized baggage-sharing marketplace which establishes the connection between the individuals looking for the cheaper, economical and fast domestic and across the border shipping operations with the travelers who have extra luggage space, making a suitable situation for both traveler and buyer. Customer will be able to

search among the products listed by the traveler which are useless for him (buying them at less cost as compared to the markets) and also allow buyer to request a specific item from the traveler traveling along a specific route.

The platform will allow travelers to:

- List their travel plans and available luggage capacity.
- Post unwanted items for sale that only add few kilograms to thier baggage.
- Access an AI chatbot for real-time destination tips, like local markets or cultural and destination insights.

Buyer will be able to:

- Browse traveler listings to find available luggage space for delivery.
- Request delivery of parcels cutting down unnecessary shipment cost and headache.
- Purchase items listed for sale by travelers at a lower cost and better conditions.

This platform resolves the common issues of international shipping which includes custom clearance, heavy duty structures and waiting delays by enabling travelers to utilize thier luggage capacity at its maximum and earn extra money, while buyers gain from a decreased dependency on traditional courier services and lower import taxes, delivery time and headache.

In short, this project focuses on building a working prototype that demonstrates how a digital baggage-sharing marketplace can function. It's about proving the idea, showing its potential, and laying the foundation for future expansions.

2.6 Sustainable Development Goal (SDG)

Following SDGs align with our project.

2.6.1 SDG 8: Decent Work and Economic Growth

Our project aligns with SDG 8: Decent Work and Economic Growth by creating new income opportunities for travelers who would otherwise leave their luggage space unused. Instead of wasted capacity, travelers can earn money while helping others access cheaper deliveries. As shown in figure 2.1

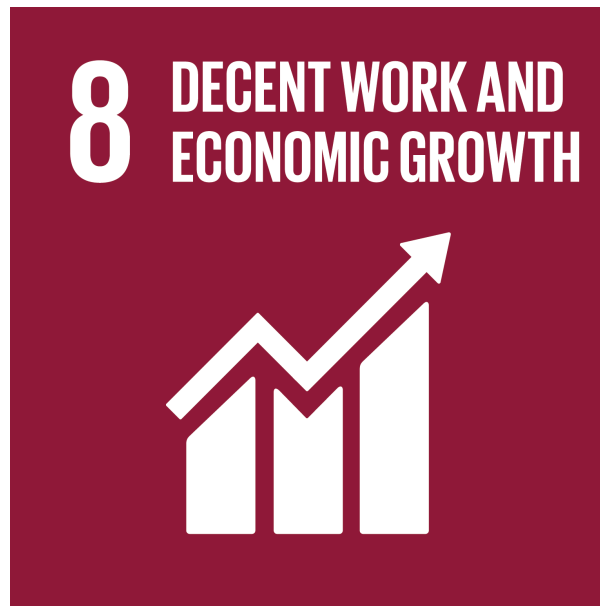


Figure 2.1: Decent Work and Economic Growth

2.6.2 SDG 12: Responsible Consumption and Production

Additionally, project indirectly aligns with SDG 12: Responsible Consumption and Production because it encourages people to make better use of existing travel capacity and even resell unwanted items locally instead of discarding them. As shown in figure 2.2



Figure 2.2: Responsible Consumption and Production

2.7 Constraints

ShareGo relies heavily on travelers willing to participate and on buyers trusting the system. That means our platform's success depends on reaching enough users to create a reliable marketplace. Also, we cannot control customs regulations or legal restrictions for cross-border shipping as these vary widely and must be managed by users.

2.8 Business Opportunity

Inefficiency of traditional shipping methods have proved that there is a need for secured, fast and reasonably priced international shipping particularly for smaller packages. Traditional courier services are often very costly, slow and complex. Thus, by turning travelers into unofficial couriers, ShareGo offers financial advantages. Customers receive quicker, less expensive deliveries, and travelers can make profit from extra luggage space. Additional growth opportunities include the possibility of branching out into local item sales and travel assistance.

2.9 Stakeholders Description/ User Characteristics

We can group the ShareGo users as:

- Travelers: People who travel frequently and have extra luggage space. They want to earn some money by carrying goods for others and may also want to sell unwanted items locally before their journey.
- Buyers: Individuals who want cheaper, secure and flexible alternative to traditional couriers and faster delivery options.
- Admin: The bridge between the travelers and buyers to ensure safe and smooth operations.

2.9.1 Stakeholders Summary

Table 2.1: Stakeholders Summary

This table identifies the main stakeholder groups for the ShareGo platform, their roles, and key interests.

Stakeholder Group	Role	Key Interest
Travelers	Provide luggage space and sell unwanted items	Generating extra income
Buyers	Request shipment or purchase items	Affordable and reliable delivery
Platform Administrators	Manage the platform, user support, and security	Secure and smooth operation

2.9.2 Key High-Level Goals and Problems of Stakeholders

Following are the things that matter to the stakeholders:

2.9.2.1 Travelers

- Goal: They want to generate an extra income, make travel easier by offloading unwanted items.
- Problem: Need a secure and trustworthy system to avoid fraud or disputes.

2.9.2.2 Buyers

- Goal: Access affordable, faster shipping while avoid heavy import taxes and unnecessary custom headache.
- Problem: Risk of delays, damaged items, or unreliable travelers

2.9.2.3 Administrator

- Goal: Keep the platform safe, fair, and legally compliant.
- Problem: Ensuring trust, resolving conflicts, and managing operations within resource limits.

2.10 Conclusion

In short, chapter 2 offers a thorough understanding of the problem domain. It points out that costly, slow shipping and wasted luggage space are problems that must be resolved. ShareGo aims to fill that gap while making travel more enjoyable and shipping less expensive by connecting buyers with travelers through a digital platform. Therefore, the main objective is to develop something practical, secure, and user-friendly. There is real business potential, especially for travelers and consumers finding secure and affordable solutions, even though constraints like customs laws and limited resources keep things in perspective.

Chapter 3 Literature Review / Related Work

Before building ShareGo, we should look that how others have tried to solve similar problems. This section walks through the main research on crowdshipping, crowd logistics, and sharing economy platforms. The goal is to understand what have been already done, what work has been done, and where the gaps still are.

3.1 Definitions, Acronyms, and Abbreviations

Some of the important acronyms, abbreviations, and definitions are given below:

AI: Artificial Intelligence

P2P models: Peer to peer models

et al. : and others

3.2 Detailed Literature Review

In this section we will reviewing the research done before by different researcher and the similar applications that are developed to solve user problems.

3.2.1 Public Transport as a Platform for Crowd shipping

Fessler et al.[1] studied the behavior of metro users and found about 40 percent would be up for carrying parcels if the pay was right and the packages weren't too heavy. Younger commuters were most interested. They founded that people are open to it, but things like coordination about delivery point and safety of the parcel could make or break the idea.

Serafini et al.[2] carried out a similar experiment in Rome. Their survey showed an even higher willingness (up to 86 percent) when lockers were placed conveniently in metro stations to ensure the safety of parcels so that the required traveler can take it or place the parcel in locker. Young individuals were flexible in it but want to have a good financial benefit. Similar to Fessler's research, it only tells us what people claim they would do, not what they really do because it was not practically tested.

While taking crowd shipping in view Cebeci et al.[3] conducted a study that whether crowdshipping actually replaces trips or creates new ones. In the Netherlands, they found most people would deliver during existing trips, but low-income respondents might make new journeys for extra income, potentially canceling out environmental gains.

This shows that the crowdshipping on public transportation or idea of making travelers as courier person can be successful as long as it aligns with the scheduled routes of traveler and makes commuting

convenient. Although it is difficult to setup but once it is functional, it has a great potential.

3.2.1.1 Relevance with ShareGo

This aligns perfectly with ShareGo's business model, which involves travelers using their already planned trips to move packages and to generate extra income while on their trips without any headache.

3.2.2 Crowdshipping for Sustainable Urban Logistics

To understand in more depth Mohri et al.[4] compiled 96 studies. They found out that when done correctly, crowdshipping consistently lowers expenses and emissions, particularly when paired with technology like matching algorithms for routes and parcel lockers to ensure thier safety. The problem is that most studies rely on simulations rather than actual trials, and trust and fairness are still weak points. Alan Mckinnon [5] looked at early crowdshipping platforms especially P2P platforms. He found that it is a good and better approach but the startup failure rate is very high. Theft, unreliable couriers, and poor route matching can quickly kill the benefits. Because the main factor which ensures success is parcel safety and security. Also the route should match the traveler so that it can facilitate him instead of frustrating him.

In short, the research community loves the idea of rowdshipping, but the execution gap is real.

3.2.2.1 Relevance with ShareGo

ShareGo targets that gap directly and efficiently by smart and secure handling of a parcel by the traveler, a verified-traveler system to ensure traveler and user benefits safety, all of which these studies claimed were lacking.

3.2.3 Social Crowd Logistics: Turning Ordinary People into the Supply Chain

Mladenow et al.[6] in their research found that the social crowds can manage data collection and delivery if there is a secure digital platform. Although their research highlights the trust and legal concerns, it also found the flexibility and cost benefits that it can provide if there is a secure platform for social crowds.

Later on, Mladenow et al.[7] performed research and analyzed various crowdsourcing P2P platforms.. It shows that people want to contribute if given the proper mobile tools and incentives. While maintaining security and quality are the difficulty.

According to various projects examined by Carbone et al.[8] crowd logistics can significantly reduce costs and add value in every case whether it is forwarding, delivery, or storage. But most of the projects are still on small scale and operating in unclear legal areas.

It clearly shows that crowd logistics is effective when technology help people by ensuring trust and safety.

3.2.3.1 Relevance with ShareGo

These findings align with ShareGo's principle. By ensuring trust and safety for users, and allowing travelers to earn income by delivering parcels, ShareGo shows strong potential.

3.2.4 Sharing Economy Principles in Logistics

These ideas could transform supply chains through shared warehouses, pooled transportation, and digital coordination for more efficient asset use, according to DHL's Sharing Economy Logistics report [9]. Using actual cases, such as IKEA's shared logistics models, Ocicka and Wieteska [10] elaborated on that, demonstrating how cooperation and digital tools promote cost effective sustainability.

They both emphasize that collaboration is only successful when openness, trust and transparency are present, which is not present in most of the logistics startups.

3.2.4.1 Relevance with ShareGo

With the support of smart matching, verification, and transparent communication, ShareGo is an example of applying sharing economy principles to logistics, converting unused luggage space into a shared resource.

3.3 Related Applications

3.3.1 Grabr

One of the earliest international travel-based delivery services was Grabr [11]. It links tourists who are going there with those who wish to purchase goods from overseas. Travelers make offers after a user posts a request, and the app withholds payment until the package is delivered safely. The idea is rather than making new trips, make use of existing ones. Grabr's greatest strength is its global presence, but it also has to deal with issues like trust, customs, and delivery dependability, which ShareGo seeks to resolve through verification.

3.3.2 PiggyBee

PiggyBee [12] is also a P2P platform that allows traveler to generate extra income along with their trips. By carrying goods requested by others it allows customers to get the product from other countries and allows travelers to facilitate other people, generating a handsome amount of money. It is based on

mutual convenience and goodwill travelers assist others in obtaining goods that are unavailable locally. The model demonstrated that if the platform maintains accountability and transparency, users will be willing to trust strangers.

3.3.3 Roadie

Roadie [13] applies the same P2P delivery logic to local transportation. It connects regular drivers with local delivery needs like groceries, furniture, and even pets, saving unnecessary travel. It is not only about saving money and to help people traveling from one point to another, it is about using the capacity already on the road to reduce traffic on roads, pollution and to properly use available resources.

3.3.4 Nimer

Nimer [14] focuses on local and regional deliveries. Items that need to be moved at a specific destination are posted by users, and for a nominal fee, someone who is already traveling that direction can volunteer to take them and get financial benefits along his already scheduled route. It uses empty car space and is based on the idea that "someone is already going there." Nimer was accepted by the people because it highlights the importance of already matching routes and ease of use between senders and couriers.

3.3.5 Airfrov

Airfrov [15] was a Singaporean startup. It allowed users to request goods from abroad and paired them with tourists who could deliver them back. Both parties were rated after payments were safely held until delivery. Up until the pandemic struck and international travel stalled, the idea was successful throughout Southeast Asia. Nevertheless, Airfrov demonstrated that there is a significant need for traveler-based delivery in areas where product access is restricted.

3.3.6 Fiverr

Even though Fiverr [16] focuses on digital freelancing rather than logistics, the marketplace mechanics are the same. Others purchase the services that people list as being available. Secure payments, public reviews, and verified profiles power the system. It shows that secure, transparent and easy to use platform can establish trust between strangers, which is exactly what ShareGo will be doing.

3.3.7 TaskRabbit

TaskRabbit [17] matches local "Taskers" with people need help in physical tasks like moving furniture and other daily routine tasks. It is P2P, based on location and relies on dependability and reputation.

It rely on regular people to carry out practical tasks in a safe and effective manner, with technology managing matching and payment.

3.3.8 Uber

Uber [18] changed the people thoughts about travel and transportation. It links drivers and passengers in the area. There are no middlemen or bus stations involved. Uber's P2P platform structure, which is based on convenience, trust, and real-time location, is what makes it relevant in this context. Similarly ShareGo is based on the same idea of fulfilling people demands by making use of the human capacity that is currently available similary as uber do in the form of driver. Allowing driver to generate income and allow user to travel with ease.

3.4 Literature Review Summary Table

This section contains a compact summary of the literature reviews we carried out. First table summarizes the research articles. Second table is of the various similar applications.

Table 3.1: Literature Review Summary Table

author	methods	results	Limitations
DHL Trend Research [9]	Case study analysis; industry reports	Sharing cut off idle capacity, costs	Case studies may overestimate adoption; legal variations
Fessler et al.[1]	Survey of various commuters; logit models	Young users favor crowdshipping with pay	Hypothetical data; untested operational feasibility
Mladenow, Bauer, Strauss [6]	Literature review and platform analysis	crowd logistics cuts costs, boosts engagement	lacks data, trust and privacy issues
A. Mckinnon [5]	Market analysis	Crowdshipping lowers costs, emissions	scalability, reliability concerns
Mohri et al.[4]	Review of 96 studies; content analysis	Cuts costs, emissions; algorithms, lockers help for security	security issues, limited real world data
Mladenow, Bauer, Strauss [7]	Literature review; platform analysis	Crowdsourcing saves costs, aids sustainability	Limited data; security, legal challenges
Carbone, Rouquet, Roussat [8]	Case study of 57 initiatives; web data	Flexible and cost efficiency	experimental, scalability issues
Ocicka, Wieteska[10]	Desk research; case studies (e.g IKEA model)	Sharing boosts efficiency, sustainability	Lack of trust and transparency.
Serafini, Nigro, Gatta [2]	Survey of metro users; logit models	Willing people if good benefits	Hypothetical data; operational issues untested
Cebeci et al.[3]	Survey from people and travelers	Most use existing trips; lockers boost acceptance	Hypothetical data, untested operations

Table 3.2: Related Applications Summary Table

Application	Feature	Relevance to Application	Limitations
Grabr [11]	cross border delivery; travelers bring items for buyers; secure payment escrow	Traveler based logistics model; promotes trust and global sharing	Limited to international deliveries
PiggyBee [12]	Community driven crowdshipping; reward based; user reputation system	Encourages sustainable, social sharing economy	Smaller user base; limited geographic coverage
Roadie [13]	Crowdsourced local delivery within U.S; route based item pickup	Demonstrates large scale implementation of logistics; real world validation of concept	Restricted mainly to the U.S; focused on parcels, not flexible user to user sharing
Nimber [14]	Connects senders and bringers; focuses on sustainability; app-based coordination	trip-based item transport	Service availability limited to certain regions; smaller active community
Airfrov [15]	travel based product shopping; escrow payment; user verification and ratings	travelers delivering products across borders	Service discontinued; relied heavily on international travel
Fiverr [16]	Freelance services marketplace; gig based model; secure payments	service listing and transaction model	Focused on digital services, not logistics; less relevant to physical delivery
TaskRabbit [17]	Local service outsourcing; connects users with nearby helpers for physical tasks	trust based local matching system	Limited to certain regions; more labor-oriented than travel-based
Uber [18]	On-demand ride-hailing and delivery; real time tracking	matching logic, trust model, and scalability potential	Focused on rides and food delivery; requires heavy operational infrastructure

3.5 Conclusion

In this chapter we looked at various studies and practical applications that will influence shared logistics in the future. Using studies on crowdshipping based on public transit and global platforms like Grabr, PiggyBee, and TaskRabbit, the review illustrated how community involvement and digital tools are transforming last mile delivery. Though they still worried about the problems like coordination, trust, transparency, security and regulation, existing models show that people are willing to participate in shared delivery systems. These observations formed the foundation of ShareGo, which implement the lessons learned from earlier attempts to create a system that is open, dependable, and truly sustainable.

Chapter 4 Software Requirement Specifications

This chapter contains all the important features of our product. It contains functional and non-functional requirements. It also contains GUI samples, use cases, and database design.

4.1 List of Features

Important features of our system are given below:-

- It can register new users.
- Both travelers and senders can manage their personal details.
- Travelers can post their travel dates, routes, and available luggage capacity.
- Users can book the traveler according to their suitable dates, luggage capacity and routes of traveler.
- Travelers can list local goods (e.g., crafts, souvenirs) for sale at their destination.
- Built in escrow holds payments until delivery is confirmed ensuring payment safety for both user and traveler.
- An AI chatbot to provide travelers with destination insights and local shopping tips.
- Admin control over the platform to monitor transactions and to make sure the platform runs smoothly.

4.2 Functional Requirements

Functional requirements of all the users are given below.

4.2.1 Functional Requirements for Travelers

- Traveler can create and manage its account.
- Traveler can list his travel details (dates, route, available capacity) and the fees he will charge.
- They can view buyers request.
- Travelers can communicate with buyers to confirm details or adjust pricing.
- Traveler can list items for sale getting rid of unwanted items.
- Travelers get paid via escrow after cutting suitable platform fee.

- Traveler can use AI chatbot to get destination insights.

4.2.2 Functional Requirements for Buyers

- Buyers can create and manage their accounts .
- Buyer can view goods listed by travelers and can purchase them at cheaper prices as compared to the market rate.
- Buyer can ask the item from the traveler to bring according to its suitable date and route.
- Buyers can communicate with the travelers.
- Buyer will pay via escrow, holding the payment until the traveler confirms delivery.

4.2.3 Functional Requirements for Admin

- Admin can view, edit or delete the user account.
- Admin make sure that platform run smoothly.

4.3 Quality Attributes

ShareGo main objective is to create a reliable and secure platform that connects the travelers with the buyers requiring affordable delivery options. These quality attributes make sure the platform feels reliable, fast, and easy to use for both travelers and buyers. Quality attributes are given below :-

- Reusability
- Security
- Efficiency
- User Experience
- Maintainability
- Responsive

These are the quality attributes which makes sure that our platform is comfortable for both travelers and buyers.

4.4 Non-Functional Requirements

Non functional requirements are given below.

4.4.1 Performance

- The system should not lag.
- Page loading time should be under 3 seconds .
- Actions like payments transactions etc. should be done under few seconds.

4.4.2 Reliability

The platform should not experience crash and data loss. The system must accurately and instantly save the data when a traveler accepts a delivery or a buyer posts.

4.4.3 Security

The system should be secure. It should prevent data loss and unauthorized accesses to maintain the data privacy of its users.

4.4.4 Flexibility

The system should work every time without any problems. It should easily adapt changes according to the changing needs.

4.4.5 Usability

The platform should be interactive and easy to use so the person who did not know so much about technology feel comfortable while using application.

4.4.6 Scalability

The system should allow new users to register and work as the platforms grow. It should handle the data without crashing.

4.4.7 Data Integrity

Every transaction, message, and update should be accurate and consistent.

4.4.8 Availability

System should be available every time user access it if the maintenance is going on then user should be notified.

4.5 Assumptions

Following assumptions are made while making ShareGo:-

- Users should have stable internet connection.
- User should have a device compatible with the application.
- Travel will enter accurate travel details and will demand suitable fees for carrying items.
- Buyer will request a reasonable item that can be carried by the user.
- Users should know basics of application before using it.
- Payments and API operations will function properly.

4.6 Use Cases

This section lists some of the use cases of our project.

4.6.1 User Login

Name	User Login		
Actors	User		
Summary	The user will enter required credentials and after verification he will be taken to the home page.		
Pre-Conditions	The user must not be logged in first and have already created account.		
Post-Conditions	The user is allowed to access after login and taken to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user opens the login page.	2	The login page is shown and ask about user credentials.
3	The user enters valid credentials.	4	After verification user logged in to the home page.
Alternative Flow			
5	The user enters invalid credentials.	4-A	The system responds with an error.

4.6.2 User Sign up

Name		User Sign up	
Actors		User	
Summary		The user will enter required credentials and after verification he will be registered and taken to the home page.	
Pre-Conditions		The user is not logged in and signing up for the first time.	
Post-Conditions		The user is allowed to access after sign up and taken to the home page.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user opens the signup page.	2	The signup page is shown and ask about user credentials.
3	The user enters valid credentials.	4	The user is registered and is taken to the home page.
Alternative Flow			
5	The user enters invalid credentials.	4-A	The system responds with an error.

4.6.3 User Logout

Name		User Logout	
Actors		Users who have already logged in	
Summary		Login user will be able to loge out.	
Pre-Conditions		The user must logged in.	
Post-Conditions		The user is logged out.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user clicks the log out.	2	User logged out login page showing.

4.6.4 Available Luggage Capacity Listing

Name	Available Luggage Capacity Listing		
Actors	Traveler/ user who is traveling.		
Summary	Allow traveler to enter its route, available luggage capacity, budget and fees he will charge.		
Pre-Conditions	Traveler must logged in.		
Post-Conditions	Traveler luggage capacity details are listed.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	Traveler click on the listing button.	2	System displays the available luggage capacity details listing page.
3	Traveler provide the required details for listing.	4	The system saves the information.
5	User click the add button.	6	The luggage capacity details listed.
Alternative Flow			
7	User did not enter all details and clicked add.	6-A	The system responds with an error.

4.6.5 Luggage Capacity Listing Deletion

Name	Luggage Capacity Listing Deletion		
Actors	Traveler who has listed luggage capacity.		
Summary	Allow the traveler to delete the luggage capacity listing.		
Pre-Conditions	Traveler must logged in.		
Post-Conditions	Luggage capacity details is deleted.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	User nevigatate to the luggage listing.	2	System displays listing page along with delete button.
3	User clicked the delete button.	4	The listing is deleted.

4.6.6 Luggage Capacity Listing Updation

Name		Luggage Capacity Listing Updation	
Actors		Traveler who has listed luggage capacity.	
Summary		Allow the traveler to update the luggage capacity listing.	
Pre-Conditions		Traveler must logged in.	
Post-Conditions		Luggage capacity details is updated.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User navigate to the luggage listing.	2	System displays listing page along with update button.
3	User clicked the update button.	4	Listing updation form displayed
5	User clicked the save button.	6	Updated luggage listing saved
Alternative Flow			
7	User did not enter properly and clicked save.	6-A	The system responds with an error.

4.6.7 Marketplace Listing

Name		Marketplace Listing	
Actors		User	
Summary		Allow traveler to list the unwanted items.	
Pre-Conditions		User must logged in.	
Post-Conditions		Item is listed.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	Traveler click on the listing button.	2	System displays the item listing page.
3	User provide the required details for listing.	4	The system saves the information.
5	User click the add button.	6	The item details are listed.
Alternative Flow			
7	User did not enter all details and clicked add.	6-A	The system responds with an error.

4.6.8 Marketplace Listing Deletion

Name	Marketplace Listing Deletion		
Actors	User who has listed the item.		
Summary	Allow user to delete the item listing		
Pre-Conditions	User must logged in.		
Post-Conditions	Item listing is deleted.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	User navigate to the item listing.	2	System displays listing page along with delete button.
3	User clicked the delete button.	4	The listing is deleted.

4.6.9 Marketplace Listing Updation

Name		Marketplace Listing Updation	
Actors		User who has listed item.	
Summary		Allow the user to update listed item details.	
Pre-Conditions		user must logged in.	
Post-Conditions		Item details are updated.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User nevigatte to the item listing.	2	System displays listing page along with update button.
3	User clicked the update button.	4	Listing updation form displayed
5	User clicked the save button.	6	Item updated details saved
Alternative Flow			
7	User did not enter properly and clicked save.	6-A	The system responds with an error.

4.6.10 Profile View

Name		User profile view	
Actors		User	
Summary		User can view their information in profile.	
Pre-Conditions		user must logged in.	
Post-Conditions		user view profile details	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User navigate to the profile view section.	2	System displays user information.

4.6.11 Profile Updation

Name		User profile updation	
Actors		User	
Summary		User can update their profile.	
Pre-Conditions		user must logged in.	
Post-Conditions		user profile is updated.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User nevigatte to the profile view section.	2	System displays listing page along with update button.
3	User clicked the update button.	4	profile updation form displayed
5	User clicked the save button.	6	Item updated details saved
Alternative Flow			
7	User did not enter properly and clicked save.	6-A	The system responds with an error.

4.6.12 Profile Deletion

Name		User profile deletion	
Actors		User	
Summary		Allow user to delete profile	
Pre-Conditions		User must logged in.	
Post-Conditions		user profile deleted.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User navigate to the profile view section.	2	System displays profile view with delete button.
3	User clicked the delete button.	4	Profile is deleted.

4.6.13 User Review/ Rating

Name		User Review/ Ratting	
Actors		User	
Summary		User can review other user.	
Pre-Conditions		user must loged in.	
Post-Conditions		user review will be generated.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User go to the review page.	2	System displays the review form.
3	User clicked the submit button.	4	User review is submitted
Alternative Flow			
5	User did not enter properly and clicked submit.	4-A	The system responds with an error.

4.7 Hardware and Software Requirements

This section of the report contains the software and hardware requirements of our project.

4.7.1 Hardware Requirements

Hardware requirements of the ShareGo is simple as it is user friendly and easy to use. There should be a device which is compatible with the application.

4.7.2 Software Requirements

The app will be made using Flutter so that it can be compatible with both Android and iOS. The Gemini API will be used for AI chatbot functionality, and the SQLite database (via the FastAPI backend) we will be using to store all of the data.

4.8 Graphical User Interface

The GUI of the ShareGo is given in this section.

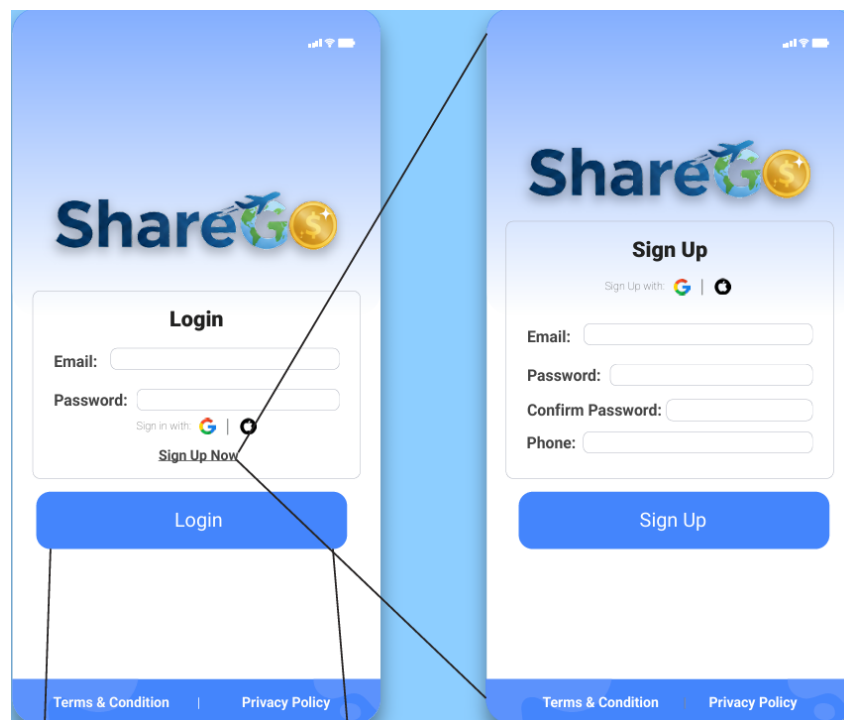


Figure 4.1: Login and Signup Page

Item Info

Title:

Product Images:
  Upload

Description:

Price:

Location

Listing Location:

List Product

Figure 4.2: Item Listing on Marketplace

Select the Date to Book Services

OCTOBER

From: To:

Counter Weight:

Humayun Raza

Hassan Gonyea

Book Now

Figure 4.3: Book Service

Luggage / Capacity Details

Available Luggage Capacity:

Pricing

Fees per kg:

Budget:

Traveler Information

Name:

Email:

Phone:

CNIC:

Exp Month: Exp Year:

Continue

Figure 4.4: Luggage Capacity Listing

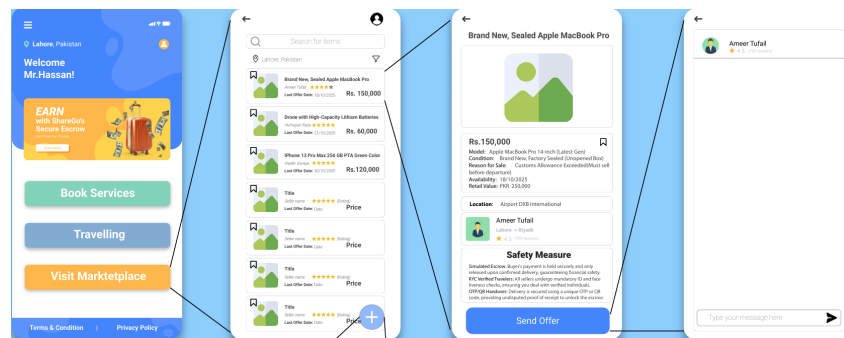


Figure 4.5: Marketplace

4.9 Database Design

4.9.1 ER Diagram

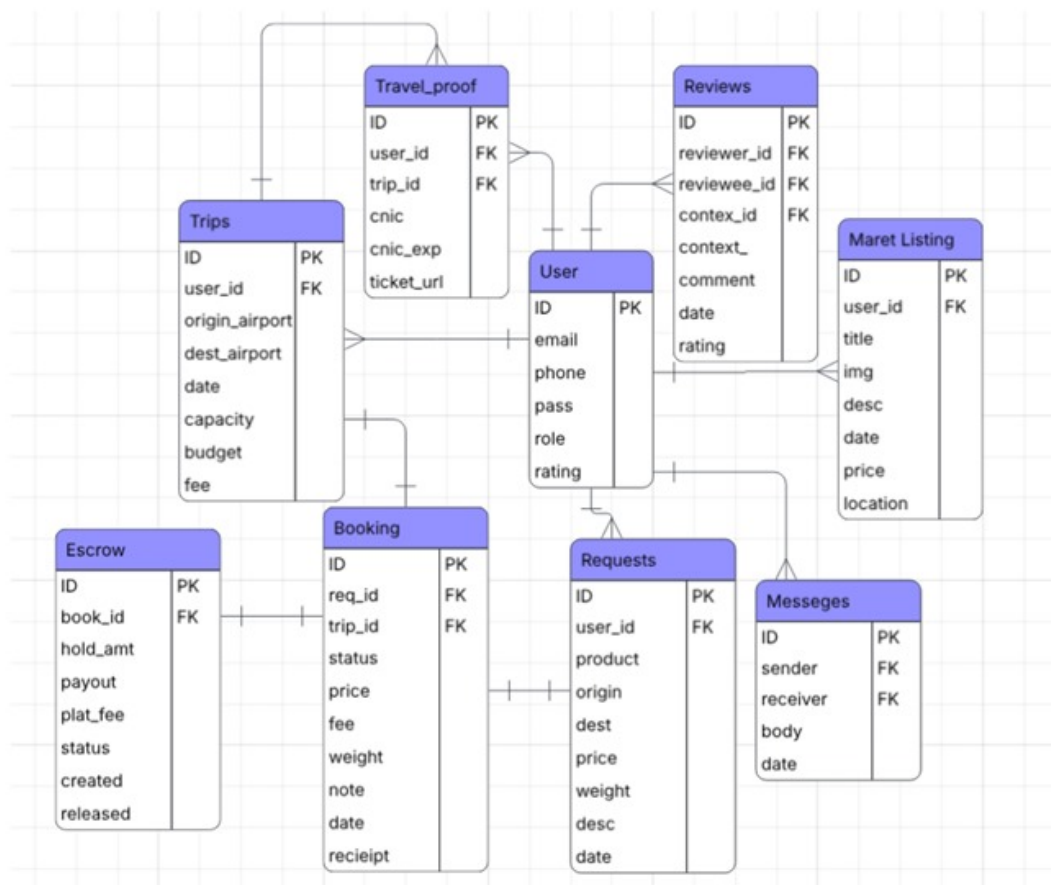


Figure 4.6: ER Diagram

Figure 4.6: ER Diagram

4.9.2 Data Dictionary

Data dictionary of each entity given in entity relationship diagram is given below.

Table 4.1: Data Dictionary of user

Entity	Attribute	Data Type	Relation To	Nullable	Description
Users	id	int		no	Unique Id of each user (PK)
	email	string		no	email of user
	phone	string		no	Phone number of user
	role	string		no	role of user
	pass	string		no	password of user
	rating	float		no	Average rating of the user

Table 4.2: Data Dictionary of Trips

Entity	Attribute	Data Type	Relation To	Nullable	Description
trips	id	int	traveler	no	Unique Id (PK)
	user_id	int	traveler	no	user id (FK)
	origin_airport	string	traveler	no	origin airport of traveler
	dest_airport	string	traveler	no	destination airport of traveler
	date	datetime	traveler	no	date of traveling
	capacity	float	traveler	no	available luggage capacity
	budget	int	traveler	no	budget available
	fee	int	traveler	no	fee charged by traveler

Table 4.3: Data Dictionary of Travel proof

Entity	Attribute	Data Type	Relation To	Nullable	Description
Travel_proof	id	int		no	Unique Id (PK)
	user_id	int	traveler	no	user id (FK)
	trip_id	int	traveler	no	trip id (FK)
	cnic	string	traveler	no	cnic number
	cnic_exp	string	traveler	no	expiry date of cnic
	ticket_url	string	traveler	no	electronic airplane ticket of traveler

Table 4.4: Data Dictionary of Booking

Entity	Attribute	Data Type	Relation To	Nullable	Description
booking	id	int		no	Unique Id (PK)
	req_id	int		no	requests id (FK)
	trip_id	int		no	trip id (FK)
	status	bool		no	booking completed or not
	price	int		no	price of product
	fee	int		no	traveler fee
	weight	float		no	product weight
	note	text		no	any note by buyer/ requester
	date	datetime		no	date
	receipt	string		no	receipt of the booking

Table 4.5: Data Dictionary of Escrow

Entity	Attribute	Data Type	Relation To	Nullable	Description
escrow	id	int		no	Unique Id (PK)
	book_id	int		no	booking id (FK)
	hold_amt	int		no	amount held by platform
	payout	int		no	amount paid to traveler
	plat_fee	int		no	platform usage fees
	status	bool		no	payment released by escrow or not
	created	datetime		no	date when amount in escrow listed
	released	datetime		no	amount release date

Table 4.6: Data Dictionary of Requests

Entity	Attribute	Data Type	Relation To	Nullable	Description
requests	id	int		no	Unique Id (PK)
	user_id	int		no	user id (FK)
	product	string		no	product requested
	origin	string		no	from where traveler is coming
	dest	string		no	where traveler will come
	price	int		no	price
	weight	float		no	product weight
	desc	string		no	description given by user
	date	datetime		no	requesting date

Table 4.7: Data Dictionary of Messages

Entity	Attribute	Data Type	Relation To	Nullable	Description
messages	id	int		no	Unique Id (PK)
	sender	int		no	user id (FK)
	receiver	int		no	user id (FK)
	body	string		no	message written
	date	datetime		no	message date

Table 4.8: Data Dictionary of Market Listing

Entity	Attribute	Data Type	Relation To	Nullable	Description
market_listing	id	int		no	Unique Id (PK)
	user_id	int		no	user id (FK)
	title	string		no	title of product being listed
	image	image		no	product image
	desc	string		no	product description
	date	datetime		no	message date
	price	int		no	product demand price
	location	string		no	seller location

Table 4.9: Data Dictionary of Reviews

Entity	Attribute	Data Type	Relation To	Nullable	Description
reviews	id	int		no	Unique Id (PK)
	reviewer_id	int		no	user id (FK)
	reviewee_id	int		no	user id (FK)
	context	string		no	reviewing booking or marketplace
	context_id	int		no	reviewing booking or marketplace (FK)
	rating	int		no	rating given
	comment	string		yes	comments added
	date	datetime		no	messege date

4.10 Risk Analysis

4.10.1 Technical and Hardware Risk

Although the ShareGo is easy to use and has basic hardware requirements but technical issue or bug may arise that can cause problems. Also if the device is too old and not compatible with ShareGo then it will not work.

4.10.2 Data Security Risk

Since user data is involved, but weak security check and unauthorized success could expose the user informations.

4.10.3 User Adoption Risk

User may dislike the app if the interface is not simple and interactive. So user should be guided about the application and its features.

4.10.4 Performance Risk

If the large number of users access app at the same time then performance can be affected.

Chapter 5 High-Level and Low-Level Design

5.1 System Overview

Travelers can generate side income through a platform that connects travelers and buyers; who want fast delivery solution with the traveler heading towards thier way. It contains modules for user registration, requests deleivry, item listing, luggage capacity listing and managing payments. The goal is to create a secure and reliable link between both users.

5.2 Design Considerations

Following is the design consideration which should be discussed while making ShareGo. Also we would explain the complete design solution in it.

5.2.1 Assumptions and Dependencies

Describe any assumptions or dependencies regarding the software and its use. These may concern such issues as:

- User should have andriod or other device that supports ShareGo.
- User should have the stable internet connection.
- Payment gateways and Gemini API should be available.
- User should have basic knowledge about the app while using it.
- Backend depends on sqlite database, FastAPI and flutter for development.

5.2.2 General Constraints

5.2.2.1 Hardware or Software Environment

- User should have basic android device.
- Ram atleast 4 GB and iOS devices.

5.2.2.2 End-user environment

- OS version should support the application.
- Basic knowledge of how to use the application.

5.2.2.3 Availability or volatility of resources

- User should have resources to run the application.
- Device should not be defective.

5.2.2.4 Network Communication

- Internet connection should be stable.
- API response should be under 3 to 4 seconds

5.2.2.5 Performance

- Large number of users should be handled by the app.
- Database operations should be optimized for the frequent queries.

5.2.2.6 Interface/protocol environments

- Easy to use and interactive interface for navigating the system.
- Easy interface to take and display data to the user.

5.2.2.7 Security requirements

- Database protected against unauthorized access.
- Database backup should be done regularly.
- User enter valid credentials and strong password to protect his data.

5.2.2.8 Data repository and distribution requirements

- SQLite database will be used via FastAPI.

5.2.2.9 Language Constraints

- User should know basics of the english language.

5.2.3 Goals and Guidelines

Following are the goals and guidelines.

5.2.3.1 User Friendly

Our system will be user friendly and easy to use. We will follow KISS principle which means keep it simple stupid by maintaining UI/ UX minimal, attractive and easy to understand.

5.2.3.2 Security

Payment release with escrow and valid access to the database and user dashboard will ensure security.

5.2.3.3 Performance

Our system will be high in performance as it will not require unnecessary space from the user and will perform all operations with minimum cost.

5.2.3.4 Modifiable

The application will be developed with easy logic and flexible design so that the new features can be added with the time.

5.2.4 Development Methods

We will be using an Agile, iterative approach with short one or two week sprints focused on important features like listings, bookings, escrow, and messaging. Each update goes through continuous integration for backend testing and manual UI checks to keep development fast. And after every progress we will have the short meeting to discuss the done work, review the progress and to schedule the rest of the work.

5.3 System Architecture

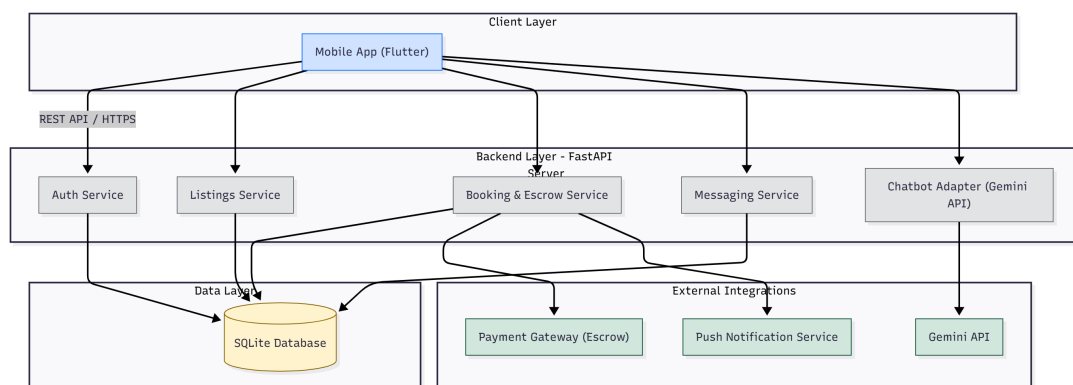


Figure 5.1: System Architecture Diagram

The system will have three layer architecture to keep everything simple and easy to manage. Client layer at the top will be managed using flutter. SQ lite database with the help of FASTAPI backend layer will control services including booking and escrow, listings, messaging, and authentication. Escrow will be used for secure payment transaction between users. External connections link to the Gemini API for chatbot interactions, provide push notifications. Because of this configuration, the system remains scalable, and simple to expand without destroying already-existing components.

5.3.1 Subsystem Architecture

As shown in following admin architecture diagram admin can manage listing, users and database. As

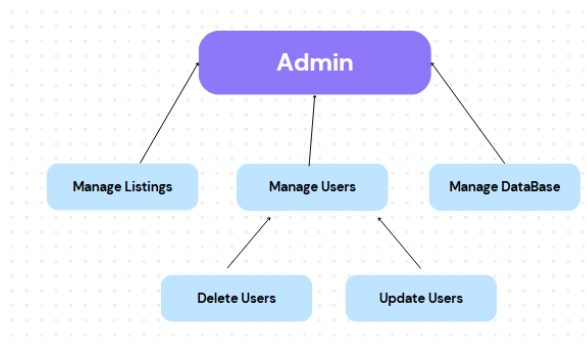


Figure 5.2: Admin Subsystem Architecture Diagram

shown in following user architecture diagram user can post all of its listing and can manage them, he can manage .

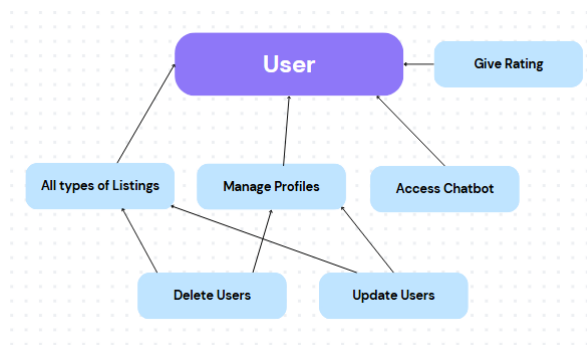


Figure 5.3: User Subsystem Architecture Diagram

5.4 Architectural Strategies

Following are the architectural strategies we will be using.

5.4.1 Technology tools (Flutter + FastAPI + SQLite)

We will be using flutter for cross-platform UI; sqlite via FastAPI for database.

- Flutter for front end as it speeds up development and provides consistent UI on Android/iOS.
- FastAPI offers fast development, automatic OpenAPI docs, and async capabilities for handling IO (e.g., external services).
- SQLite simplifies local deployment and testing during FYP.

5.4.2 Security and Trust

When you are dealing with user sensitive information and financial transactions then security cannot be compromised.

- For payment protection and to reduce frauds escrow will be used to hold the payment. Before placing booking user will handover its amount to the platform which will be released by the platform once traveler has delivered the product.
- Traveler verification will be done through its e-ticket, to confirm that whether he is traveling or not.
- Users can give ratings to others so that people can know whether that person is trustworthy or not.

5.4.3 Future Expansion

Our platform will allow to add new features and additional services as the technology evolves making ShareGo a strong, long-term, up to date and a successful platform.

5.5 Class Diagram

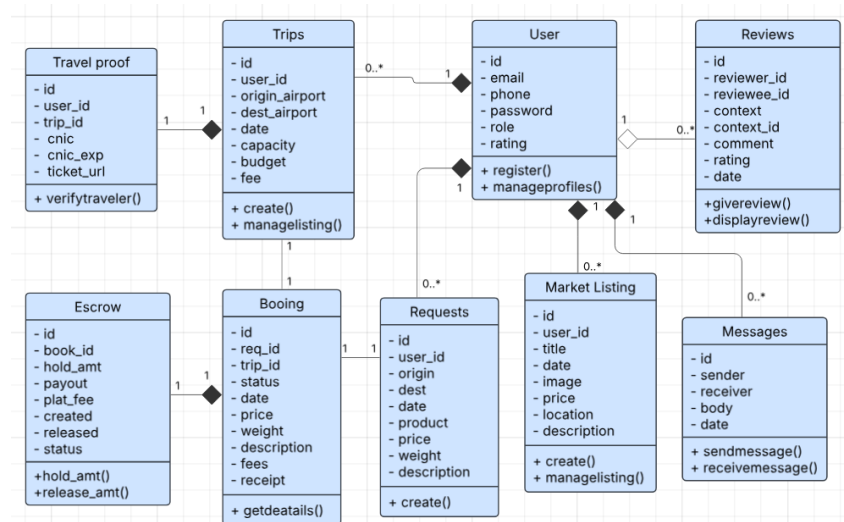


Figure 5.4: Class Diagram

5.6 Policies and Tactics

To keep the system reliable, secure, and easy to use and evolve with the technology, following policies will be followed to ensure platform efficiency and user satisfaction.

5.6.1 Coding Envoinrment

The system is built using Flutter for the frontend, providing a smooth and responsive cross-platform mobile experience. The backend is developed with FastAPI, while SQLite serves as the local database for storing user and transaction data. This ensures that the development remain efficient, and easy to scale later if needed.

5.6.2 Modular Design

Each service will be built independently so that the failure one services donnot affects the availability of whole system. This keeps the platform stable and easier to maintain.

5.6.3 System Testing

We will be using white box testing to test every buisness logic and module. Every aspect of the application will be tested to avoid any kind of error.

5.6.4 User Interface Design

Interface will be kept simple and attractive while taking user preferences in mind. Flutter's flexible UI components are used to create a consistent layout across Android and iOS. Buttons, icons and form will be placed at easy positions so that the user can nevigat through app easily.

5.6.5 Maintenance and Improvement

Once the ShareGo is completed we will keep on adding new features and and improving older once. This will make the ShareGo more compatible with the market and technologically up to date while keeping user preferences in mind.

5.6.6 Backup and Recovery

Regular database backups are maintained to protect against data loss and ensure quick recovery in case of failure.

5.7 Conclusion

The chosen architecture (Flutter + FastAPI + SQLite database) makes ShareGo simple and easy to scale with the time as number of users increases, for example switching to a cloud DB, adding more features, enable AI approvals and checking over the system to monitor user engagements. The design makes the safety possible and important through escrow and verifications, simplicity for mobile users, and modularity for maintainability and future extension.

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